Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/03/2020
PI Name:	Karma, Alain Ph.D.		
Project Title:	Computational Studies of Microstructure Formation during Alloy Solidification in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	11/16/2018	**End Date:**	11/15/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0135		
Performance Goal No.:			
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	NOTE: Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX16AB54G," with the same Principal Investigator, Dr. Alain Karma. Dendritic microstructures formed during solidification processing (such as casting, welding, and brazing) determine the properties of structural metallic alloys used in aerospace materials essential for NASA exploration in space and produced in numerous US industries with commercial applications ranging from automobile engine parts to turbine blades used in aircraft and power generation. The main project goal is to develop multiscale and multiphysics computational models to relate solidification processing conditions with alloy microstructures and properties. Those models provide enhanced capabilities to accelerate the development of advanced (light-weight, high-temperature, and extreme environment) aerospace materials produced on ground by diverse solidification processes including emerging		

Task Description:	new technologies such as additive manufacturing. In addition, they provide support for materials processing on extraterrestrial surfaces in a reduced gravity environment. Model development is carried out iteratively with microgravity solidification experiments carried out in the Materials Science Laboratory and Transparent Alloy and Device for the Study of Critical Liquids and Crystallization (DECLIC) facilities aboard the International Space Station (ISS) in collaboration with ESA (European Space Agency) and CNES (Centre national d'études spatiales, French space agency) research teams. Computationally-designed materials has been identified as a critical technology in the NASA Technology Roadmap. This research meets the general objective of this critical area of accelerating materials development by integrating physics-based computational models of materials at multiple length scales to relate solidification processing, microstructures, and properties, and by developing massively parallel implementation of computational codes on graphics processing units (computer power). The present project makes use of both phase-field (PF) simulations and the dendritic needle network (DNN) model to establish benchmark comparisons between theory and microgravity experiments. Phase-field simulations resolve the details of the dendrite tip scale, while the DNN model tracks the dynamics of dendritic networks on a much larger scale of several grains. For this reason, the DNN model has emerged as a powerful method to bridge the dendrite tip and grain scales. Those simulation tools provide the means to model microstructure formation in a diffusive growth regime over the full range of scales needed to establish quantitative benchmark comparisons with microgravity observations. Ongoing computational studies have two main objectives: DECLIC/DSI. The first objective is to model directional solidification experiments in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) onboard the ISS as part of a CNES/NASA project. Those experiments visualize for the first time in dilute transparent alloys the solid-liquid interface evolution in three dimensions under diffusive growth conditions. They provide a wealth of new observations to elucidate complex mechanisms governing the spatiotemporal evolution of three-dimensional dendritic array structures. The goal of this research is to model both DECLIC/DSI experiments and the new series of DECLIC/DSI-reflight(R) experiments. CETSOL (Columnar-to-Equiaxed Transition in Solidification Processing (CETSOL). The second objective is to model directional solidification experiments in metallic and concentrated transparent organic alloys in the framework of the ESA/NASA CETSOL-6 project (March 2018 – February 2021) carried out in collaboration with US and European partners. The general goal of this project is to improve the understanding of complex physical phenomena governing the Columnar-to-Equiaxed Transition (CET) in alloy SOLidification and its consequences for industrial castings. This project involves studies of metallic alloys and the transparent alloy system Neopentylglycol-(d)Camphor (NPG-DC) in the Transparent Alloy facility. This system adds the capability to visualize both columnar and equiaxed dendritic grain development and the CET in a concentrated alloy regime, thereby complementing the DECLIC/DSI-R studies of columnar dendrites in dilute SCN-camphor alloys.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to lightweight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales. Those tools provide enhanced capabilities to accelerate the development of advanced (light-weight, high-temperature, and extreme environment) aerospace materials produced on ground by diverse solidification processes including emerging new technologies such as additive manufacturing. In addition, they provide support for materials processing on extraterrestrial surfaces in a reduced gravity environment.
Task Progress:	The major accomplishments during the present grant period that are further described in the annual report include: • A detailed quantitative comparison of phase-field simulations experiments carried out in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) onboard the International Space Station. The results shed new light on the effect of sub-boundaries (subgrain structures) on the evolution of the primary spacing in spatially extended cellular array structures that form during directional solidification on binary alloys. Experiments and simulations were used to characterize the spatiotemporal increase and decrease of the primary spacing at divergent and convergent sub-boundaries, respectively. The primary spacing is of direct of technological relevance as it controls the mechanical properties of structural metallic alloys such as tensile strength. • The first detailed quantitative comparison of phase-field simulations and a different campaign of DSI-R experiments that access the fully developed dendritic regime using a more concentrated transparent organic succinonitrile-camphor alloy. The measured and predicted dendrite tip radius were found to be in excellent quantitative agreement for comparable primary spacings. • An exhaustive phase-field computational study of dendritic grain growth competition that enabled us to compute for the first time orientation selection maps over a complete range of grain-boundary bi-crystallography in a fully developed dendritic regime characteristic of experiments aimed at understanding the cellular to equiaxed grain structure transition during the solidification of binary alloys. Results were used to establish benchmark comparisons with the predictions of coarse-grained cellular automata finite element (CAFÉ) models in collaboration with our European partners. This study revealed the existence of different mechanisms of grain selection in opposite regimes of low and high temperature gradients.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Articles in Peer-reviewed Journals	Mota F, Bergeon N, Karma A, Trivedi R, Debierre JM. "Oscillatory-nonoscillatory transitions for inclined cellular patterns in three-dimensional directional solidification." Physical Review E. 2020 Sep;102(3):032803. https://doi.org/10.1103/PhysRevE.102.032803 , Sep-2020

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Articles in Peer-reviewed Journals	Geslin P-A, Chen C-H, Tabrizi AM, Karma A. "Dendritic needle network modeling of the Columnar-to-Equiaxed transition. Part I: Two dimensional formulation and comparison with theory." Acta Materialia. 2021 Jan 1;202:42-54. https://doi.org/10.1016/j.actamat.2020.10.009 [Note: previously reported in September 2020 as "Under review. Posted: 17 Sep 2020.] , Jan-2021
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Papers from Meeting Proceedings	Mota FL, Ji K, Lyons T, Strutzenberg LL, Trivedi R, Karma A, Bergeon N. "In situ observation of growth dynamics in DECLIC Directional Solidification Insert onboard ISS: DSI-R flight campaign." 70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019. 70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019. Paper ID: 53268. , Oct-2019